Module 5: Physical chemistry and transition elements

The content within this module assumes knowledge and understanding of the chemical concepts developed in Module 2: Foundations in chemistry and Module 3: Periodic table and energy.

This module extends the study of energy, reaction rates and equilibria, and the periodic table.

The main areas of physical chemistry studied include:

- rate equations, orders of reaction, the rate-determining step
- equilibrium constants, K_c and K_p
- acid–base equilibria including pH, K_a and buffer solutions
- lattice enthalpy and Born–Haber cycles
- entropy and free energy
- electrochemical cells.

The main areas of inorganic chemistry studied include:

- redox chemistry
- transition elements.

Synoptic assessment

This module provides a context for synoptic assessment and the subject content links strongly with the content encountered in Module 2: Foundations in chemistry and Module 3: Periodic table and energy.

- Atoms, moles and stoichiometry
- Acid and redox reactions
- Bonding and structure
- Periodicity, Group 2 and the halogens
- Enthalpy changes
- Reaction rates
- Chemical equilibrium

Knowledge and understanding of Module 2 and Module 3 will be assumed and examination questions will be set that link their content with this module and other areas of chemistry.